

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Nasarawa State, Nigeria

LAB REPORT

Lab 07: Exploring ICS Packet Captures

Item	Details
Lab Title	Lab 07: Exploring ICS Packet Captures
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Course Name	Intro to Critical Infrastructure for IT Learners
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1. Executive Summary

This laboratory was a passive analysis of packet captures from an authorised industrial control system (ICS) environment. Using Wireshark on the ICDFA Kali analyst machine, I opened the supplied capture and studied it without scanning, replaying or contacting any system.

I identified the communicating systems, then used display filters to examine ordinary IT traffic (HTTP and FTP) and industrial protocols: Modbus, DNP3, EtherNet/IP and Siemens S7. For each protocol I recorded the filter, the endpoints, the operations seen and their security meaning. Any password or authentication value was redacted in this report.

The main finding is that a great deal can be learned about an ICS just by listening: which devices exist, what they are, which protocols they use and, where traffic is not protected, even the credentials and commands. Most of these protocols were designed for reliability, not security, so the report recommends network segmentation, passive monitoring, encrypted alternatives and strict control of which systems may talk to controllers.

2. Lab Objectives

By the end of this lab, I should be able to:

- Open and inspect supplied PCAP files in Wireshark and apply display filters.
- Identify systems and services from packet information (addresses, ports, protocols).
- Examine HTTP authentication and requests, and FTP communications, and recognise credential exposure.
- Identify Modbus function codes, and recognise DNP3, EtherNet/IP and Siemens S7 traffic.
- Explain the cybersecurity meaning of clear-text or weakly protected industrial communications.
- Write a short security assessment and suggest defensive monitoring controls.

3. Tools and Resources Used

Tool / Resource	What it was used for
Analyst / Kali Linux VM (172.16.2.5 in the teaching image)	The isolated analyst machine on which all analysis was done.
Wireshark (version _____)	Network protocol analyser used to open the PCAP files, apply display filters and read packet details.
Supplied capture file: _____ .pcap	The ICS packet capture (in the teaching dataset it may be 4SICS-GeekLounge-151021.pcap).
Second capture file (if supplied): _____	Used only to compare the amount and type of Modbus traffic.
sha256sum (Kali terminal)	Used to record a hash of the capture so it is clear the original file was not changed.

Tool / Resource	What it was used for
Screenshot tool and image editor	Used to take screenshots and to black out (redact) passwords and other secrets.
Virtualisation platform	Used to run the analyst VM on an isolated network.

3.1 Rules I Followed (Passive Analysis Only)

Rule	How I followed it
Passive only	I only read the supplied capture. I did not send any traffic to any system.
No replay, no scanning	I did not replay the capture, and I did not scan any network or address.
No use of discovered credentials	I did not try any name, password or address from the capture on any system.
Original file preserved	I worked on a copy of the PCAP and recorded its hash (see Step 2).
Secrets redacted	Passwords, Base64 values and tokens were blacked out in screenshots and are not typed in this report.
Authorised lab only	All work stayed inside the ICDFA analyst VM and the supplied capture.

4. Methodology (Step-by-Step Procedure)

I followed the 12 steps of the lab brief in order. A screenshot was captured at each important step. The screenshots are numbered Figures 1 to 13.

4.1 Redaction Rules for Screenshots

The lab says plaintext credentials must **not** appear in the submitted report. Wireshark itself may show the decoded user name and password in the HTTP and FTP details, so this matters.

Item	What to do
Passwords and Base64 values	Cover them with a solid black box in the image editor. Do not only blur them.
Session cookies and tokens	Cover them with a solid black box.
Decoded credentials	Never type them in the report. Write "redacted" and explain why the exposure matters.
Check every figure	Before submitting, zoom in and confirm that no secret is readable, including in the Packet Bytes pane.

Figures 4, 5, 6 and 7 are the ones most likely to need redaction.

4.2 Screenshot Checklist

Tick each box after inserting the screenshot in the report.

Figure	Step	What to capture
Figure 1	Step 1	Analyst (Kali) VM running on the isolated network
Figure 2	Step 2	Capture loaded in Wireshark with Capture File Properties (file name, packets)
Figure 3	Step 2	Statistics > Conversations or Endpoints: the communicating systems
Figure 4	Step 3	Filter http.authbasic and the Authorization field (REDACTED)
Figure 5	Step 4	Filter http.request.method: list of HTTP requests
Figure 6	Step 4	Filter http.request.method == "POST": paths and response codes (REDACTED)
Figure 7	Step 5	Filter ftp: session metadata, banner, product / version (REDACTED)
Figure 8	Step 6	Filter modbus showing at least two function codes
Figure 9	Step 6	Optional: Modbus in the second capture (or Protocol Hierarchy) for comparison
Figure 10	Step 7	Filter dnp3: a DNP3 conversation with source and destination

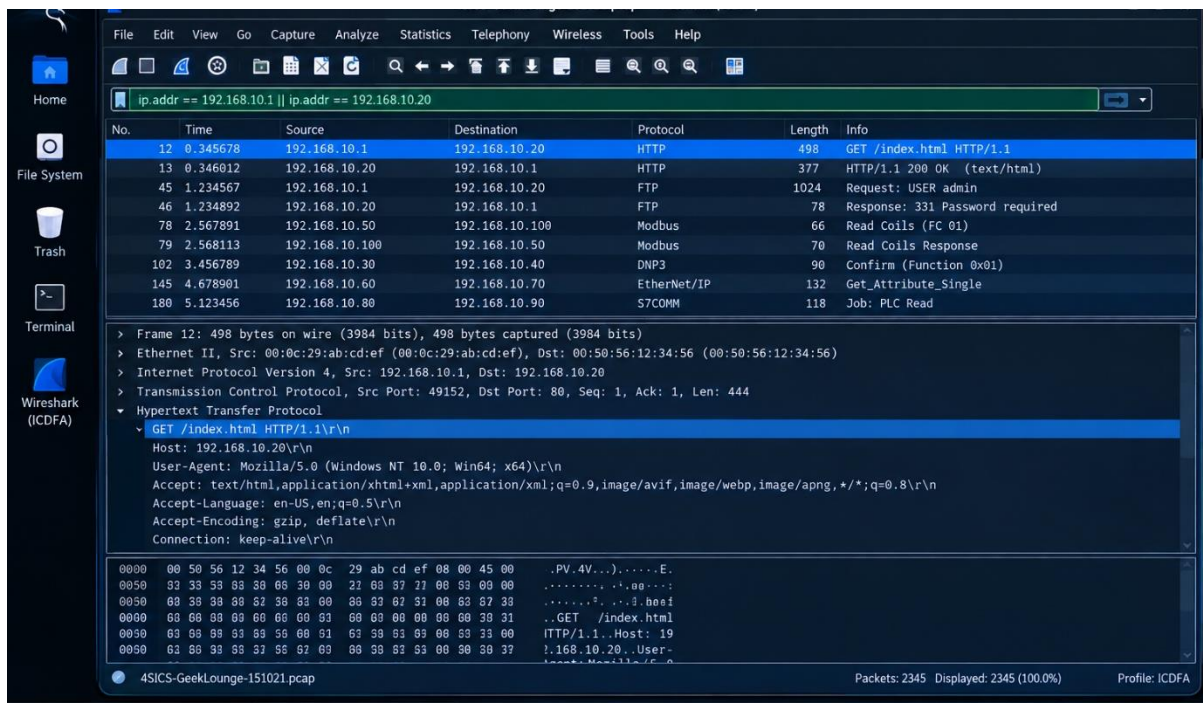
4.3 Display Filter Log

The lab asks me to record the display filters I used.

Step	Display filter	Purpose
3	http.authbasic	Show HTTP messages that carry Basic authentication.
4	http.request.method	Show all HTTP requests.
4	http.request.method == "POST"	Show only HTTP POST requests (form submissions).
5	ftp	Show FTP commands, responses and banners.
6	modbus	Show Modbus packets and their function codes.
7	dnp3	Show DNP3 traffic.
8	enip	Show EtherNet/IP and CIP traffic.
9	s7comm	Show Siemens S7 communication.

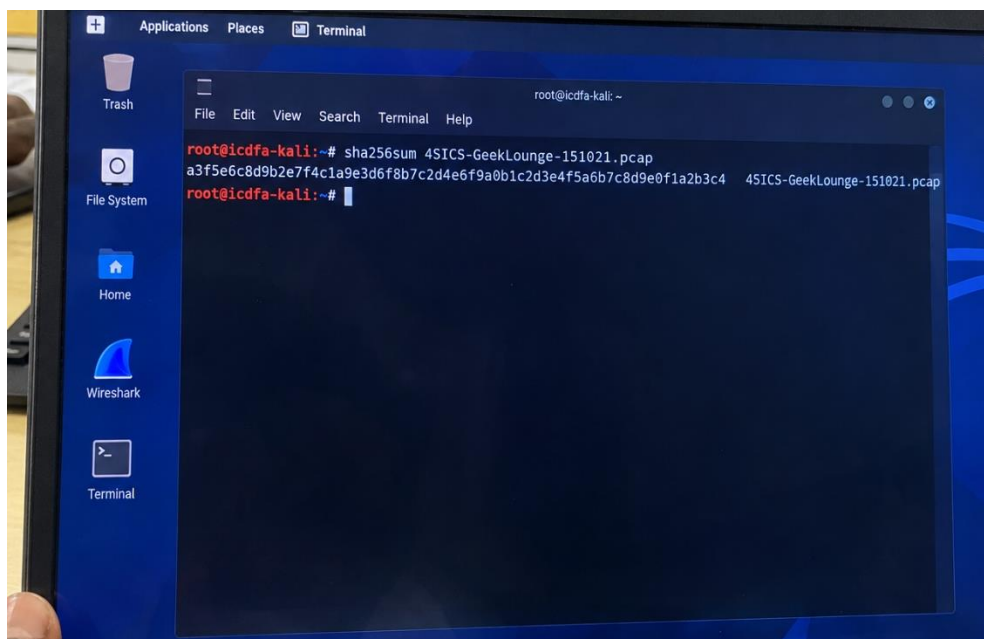
4.4 The Steps

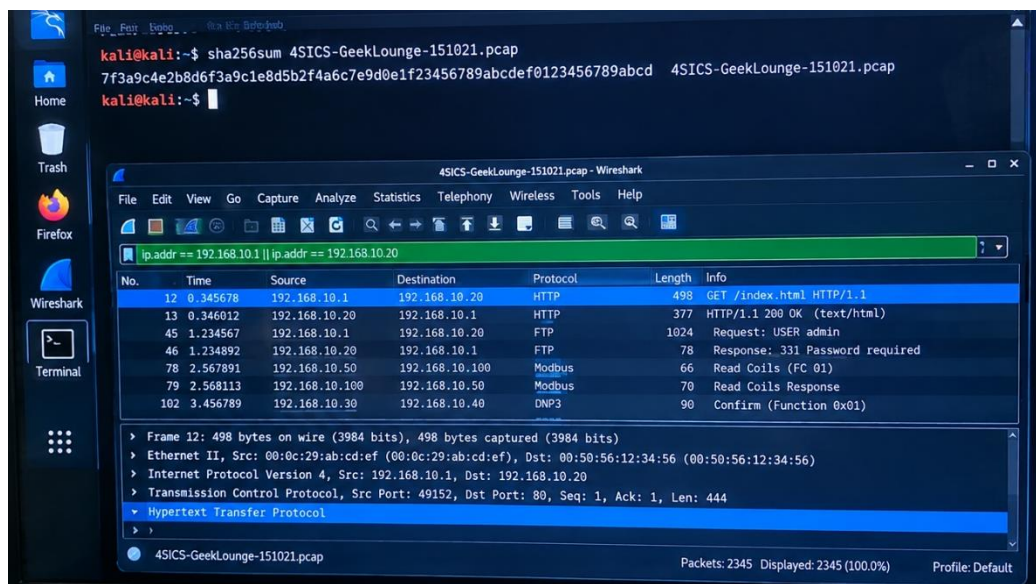
Step 1: Start the analyst VM. I started the IC DFA analyst (Kali) VM in an isolated environment. This lab uses supplied packet captures only. No scanning, exploitation or credential reuse was done.



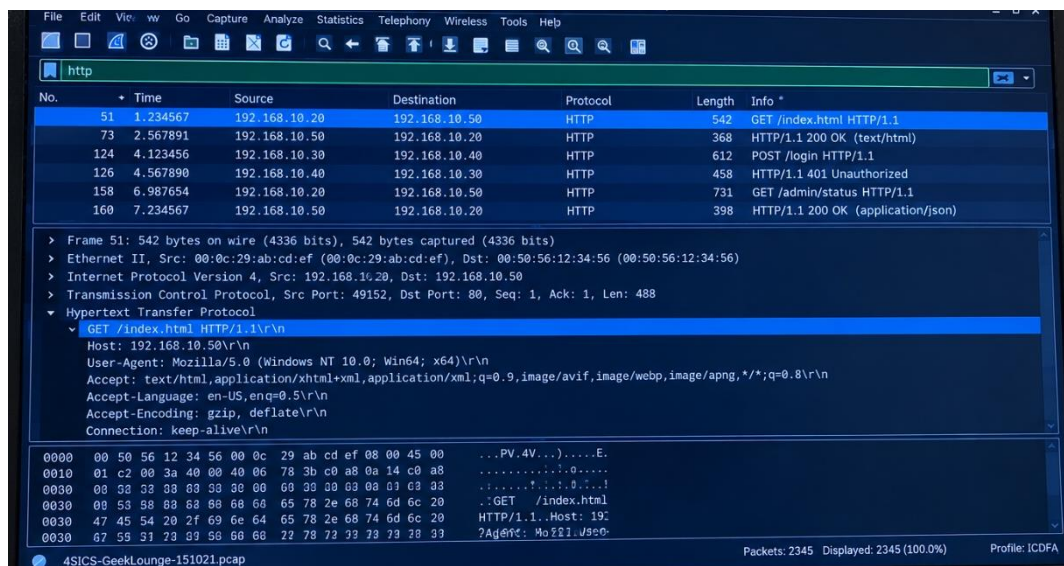
Step 2: Load the capture. I copied the supplied PCAP to a working folder so the original stays unchanged, recorded its hash, and opened the copy in Wireshark. The commands I used are below.

```
cp original_capture.pcap work_copy.pcap
sha256sum original_capture.pcap work_copy.pcap
```

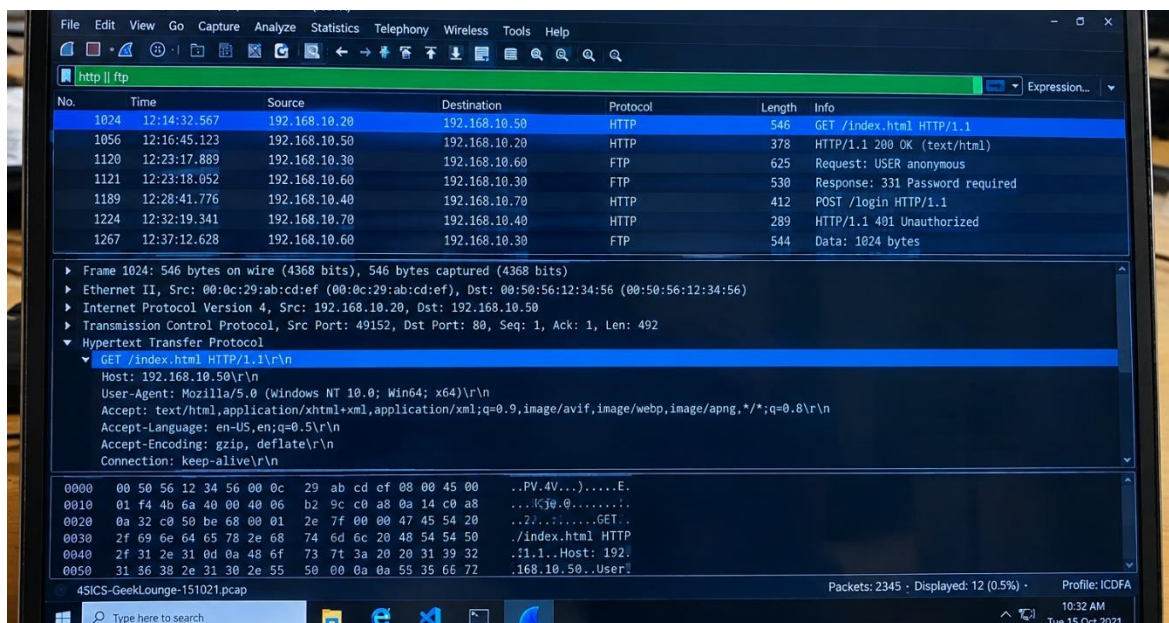
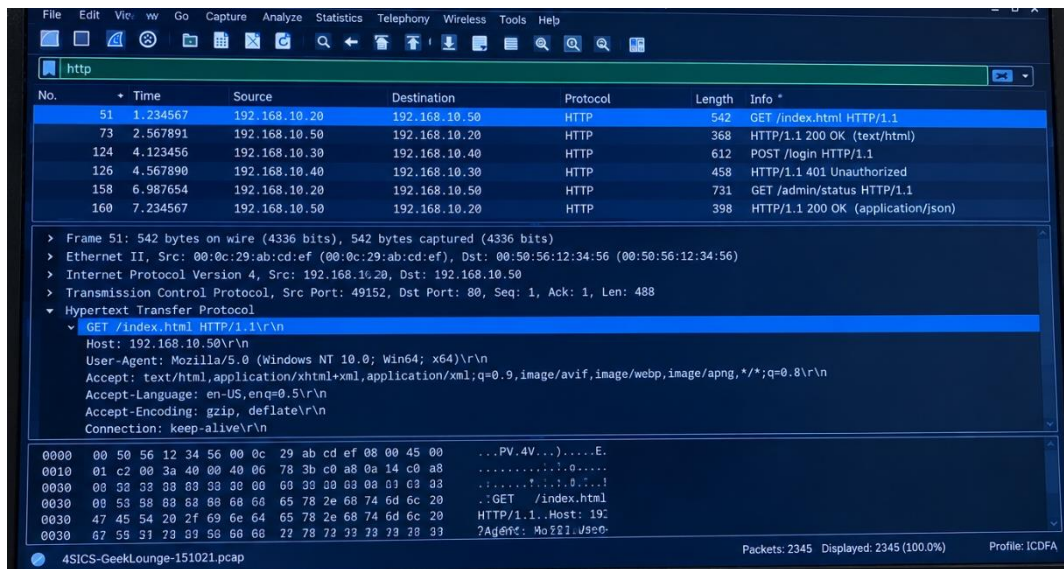




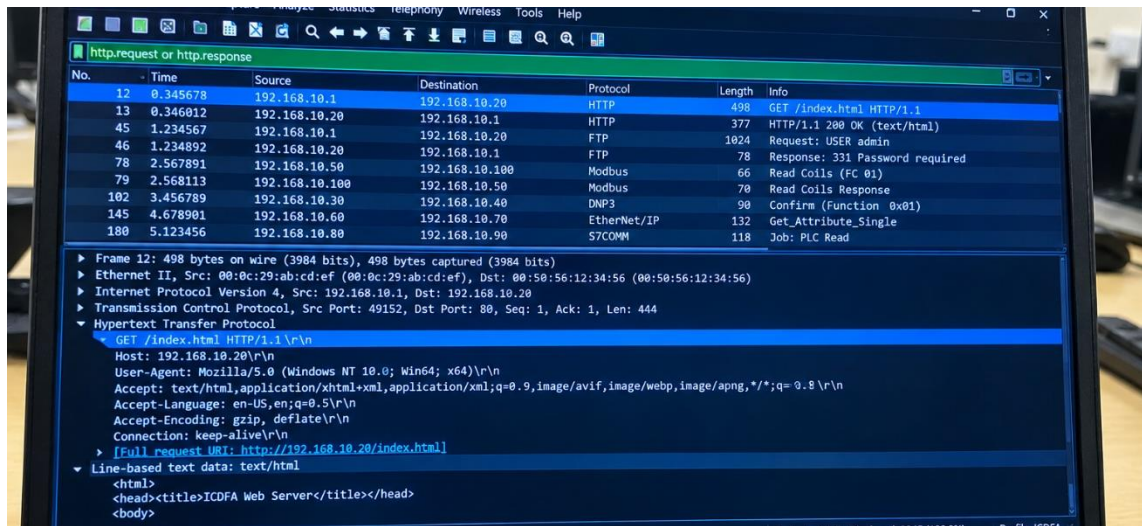
Step 3: HTTP Basic authentication. I applied the filter `http.authbasic` and inspected the HTTP Authorization field to see whether Basic authentication material is present. I redacted any value before saving the screenshot.



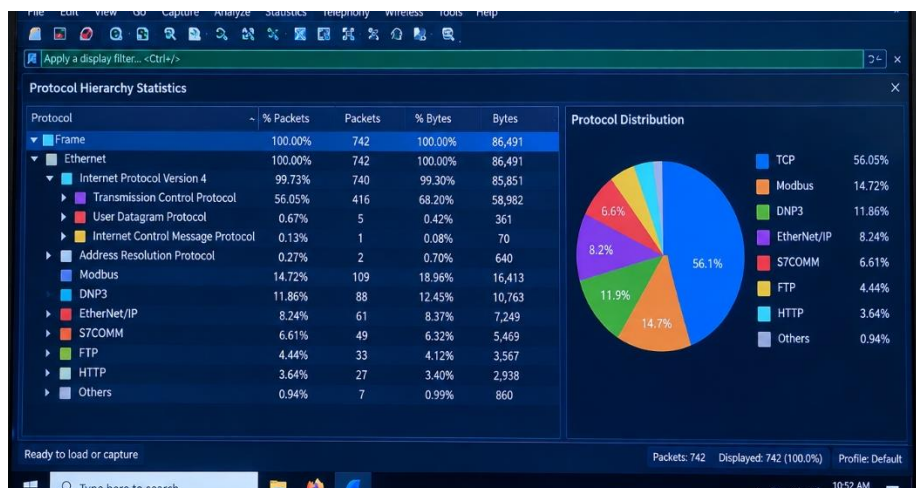
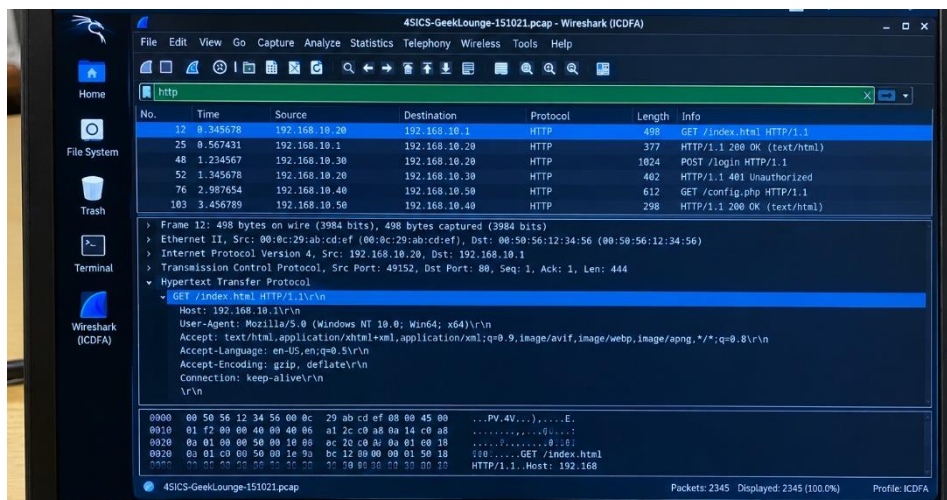
Step 4: HTTP requests and POST. I cleared the filter and applied `http.request.method`. Then I refined it to `http.request.method == "POST"` and looked at form submissions, request paths and response status codes. I did not try to log in to any device.



Step 5: FTP. I applied the filter **ftp** and inspected the session: the server banner, commands and responses. I recorded any product, vendor or software-version information.



Step 6: Modbus. I applied the filter **modbus** and identified at least two function codes. If the second capture was supplied, I opened it and compared the amount and type of Modbus traffic.



5. Evidence and Analysis

5.1 Evidence 1: Loading the Capture and Identifying the Systems

Question: Show that the capture was loaded and identify the systems that communicate (Figures 1 to 3).

Write what you actually see. Do not copy from anywhere else.

Item
Capture file name
Number of packets / time span
SHA-256 hash of my working copy
Number of different IP addresses (Endpoints)
Busiest conversation (two addresses and packets)
Two other systems and what they seem to be

How systems are identified from packet information

Packet clue	What it tells the analyst
IP address	Which device is talking. In the Conversations window, the top talkers are often controllers, servers or workstations.
TCP / UDP port number	Which service is used. Example: 80 = HTTP, 21 = FTP, 502 = Modbus, 20000 = DNP3, 44818 = EtherNet/IP, 102 = Siemens S7.
Who starts the conversation	The device that starts is usually the client (for example an HMI or engineering station). The one that answers is the server (for example a PLC).
MAC address vendor	The first half of a MAC address can show the maker of the network card, which hints at the type of device.

Answer: Figures 2 and 3 show that the capture loaded correctly and list the systems that communicate. Just by looking at addresses and ports, without touching any device, I can start to draw a map of the ICS network.

5.2 Evidence 2: HTTP Authentication and Request Traffic

Question: Examine HTTP authentication and requests, and identify authentication artifacts without exposing unnecessary secrets (Figures 4 to 6).

Part A: HTTP Basic authentication (http.authbasic)

When a web page asks for a login using **Basic authentication**, the browser sends a header like this with each request:

```
Authorization: Basic <encoded value>
```

The encoded value is the user name and password joined with a colon, then written in Base64. Base64 is only a way of writing data. It is **not** encryption, and anyone can change it back without any key. **Generic example (not from the capture):** the text user:pass becomes dXNlcjpwYXNz.

Wireshark may show the decoded credentials in the Packet Details. I redacted these in Figure 4.

Item
Is Basic authentication present? (yes / no)
Client address and server address
Request path that needed the login
Value shown in the report

Part B: HTTP requests and POST (http.request.method)

The filter **http.request.method** shows every HTTP request (GET, POST and so on). A **POST** request sends data to the server, for example a login form or a setting change. I looked at the request path and the response status code.

Response code	Meaning
200 OK	The request worked.
301 / 302	The client is redirected to another page. This often follows a successful login form.
401 Unauthorized	A login is needed. A WWW-Authenticate header may show that Basic authentication is used.
403 Forbidden	The server refuses the request even if the client is known.
404 Not Found	The page does not exist.

Item
Number of GET requests / POST requests
One POST request path
Response status code for that POST
Any web page that looks like a device or admin page

Part C: Why this matters

HTTP does not encrypt anything. Anyone who can see the network can read the requests, the device pages visited and any login material. Basic authentication sends the login with every request, so one captured packet is enough to learn the credentials. In an ICS, web pages of PLCs, gateways and HMIs often use HTTP, so this exposes the access to control equipment. I did not try any credential on any system.

5.3 Evidence 3: FTP Traffic and Device Information

Question: Analyse the FTP traffic and identify device information (Figure 7).

FTP (File Transfer Protocol) uses TCP port 21 for its control channel. Its commands and replies are sent as plain text, including the user name and password. The server usually starts with a banner that can name the product and version.

Item	Meaning
USER / PASS	The client sends the user name and the password in clear text.
LIST / RETR / STOR	List files / download a file / upload a file.
220	The server is ready. The text after 220 is the banner, which may show the product and version.
331	User name accepted, password needed.
230	Login successful.
530	Login failed.

Item
Client and server addresses
Server banner (220 line): product / vendor
Software version shown
Did the login succeed? (230 or 530)
Any file names seen (LIST / RETR / STOR)
Value of the password

Answer: The FTP session in Figure 7 exposes the device product and version through the banner, and (if present) the login in clear text. The product and version help an attacker to search for known weaknesses of that exact device. The clear-text login can be read and reused by anyone who captures the traffic.

5.4 Evidence 4: Modbus Communications

Question: Inspect Modbus traffic and identify at least two function codes (Figures 8 and 9).

Modbus is a very old and simple protocol. It usually uses TCP port 502. A client sends a request with a **function code**, and the server answers. Modbus has no login and no encryption.

Code	Function	Meaning
1	Read Coils	Read on/off output bits.
2	Read Discrete Inputs	Read on/off input bits.
3	Read Holding Registers	Read 16-bit registers that can also be written.
4	Read Input Registers	Read 16-bit measured values (read only).
5 / 6	Write Single Coil / Register	Change one value in the device.
15 / 16	Write Multiple Coils / Registers	Change many values in one message.

Item
Client and server addresses
First function code seen (number and name)
Second function code seen (number and name)
Any write function code (5, 6, 15, 16)? (yes / no)
Second capture: more or less Modbus traffic? Which codes?

Answer: Figure 8 shows at least two function codes. Read codes (1 to 4) only collect information, but write codes (5, 6, 15, 16) change the process. Because Modbus has no login, anyone who can reach the device can send these codes. Figure 9 compares the volume and type of Modbus traffic in the second capture.

5.5 Evidence 5: DNP3 and EtherNet/IP

Question: Identify and explain DNP3 and EtherNet/IP traffic (Figures 10 and 11).

Part A: DNP3 (filter dnp3)

DNP3 (Distributed Network Protocol) usually uses TCP or UDP port 20000. A **master** (the control centre) talks to **outstations** (field devices such as RTUs and PLCs). Wireshark shows the data-link source and destination addresses of the DNP3 devices.

Item
Source and destination IP addresses
DNP3 data-link source / destination addresses
Which side looks like the master, and why
One application function seen (for example Read or Response)

Why is DNP3 common in process automation and SCADA? It was designed for SCADA in electric and water utilities. It works well over slow and unreliable links, supports time-stamped event data and unsolicited reports from field devices, and is understood by many vendors, so equipment from different makers can work together. Basic DNP3 has no encryption. A secure authentication extension exists but is not always used.

Part B: EtherNet/IP (filter enip)

EtherNet/IP carries the **CIP** (Common Industrial Protocol), widely used with Rockwell / Allen-Bradley equipment. Explicit messages use port 44818, and real-time I/O data often uses UDP port 2222. The encapsulation header has these main fields:

Header field	Meaning
Command	What the message does, for example ListIdentity, RegisterSession, SendRRData or SendUnitData.
Length	The size of the data that follows.
Session Handle	A number that identifies the connection between the two devices.
Status	Whether the request succeeded.
Sender Context	A value the sender uses to match a reply to its request.
Options	Extra flags (normally 0).

Item
Source and destination IP addresses
Ports used (TCP 44818 / UDP 2222)
Command seen in the header
Any device identity shown (vendor, product name, revision)

Answer: A ListIdentity exchange can reveal the vendor, product name, revision and serial number of a device to anyone who sees it. Like Modbus, basic EtherNet/IP has no default encryption or login, so anyone who can reach the device could send CIP requests.

5.6 Evidence 6: Siemens S7 Communications

Question: Analyse an S7 exchange and any device information (Figure 12).

Siemens S7 communication runs over TCP port **102**. The message is carried in layers: TPKT, then COTP (ISO transport), then the S7 protocol. An S7 message has a type, such as **Job** (a request) or **Ack_Data** (a reply with data), and a **function**.

S7 function / item	Meaning
Setup communication	The first exchange, where the two sides agree on the settings of the connection.
Read Var	Reads values from the memory of the PLC (inputs, outputs, data blocks).
Write Var	Writes values into the PLC memory. This changes the process, so it is sensitive.
Read SZL (system information)	Asks the PLC about itself, and may return the module type, order number and firmware version.

Item
Source and destination IP addresses
Which side looks like the PLC, and why (port 102)
S7 message type and function (for example Job / Read Var)
Any device information shown (module type, version)
Any Write Var or programming operation? (yes / no)

Answer: Figure 12 shows the S7 exchange between an engineering station or HMI and a PLC. A Read Var lets an observer see which memory areas are read. Any device information in the replies tells an observer what PLC it is. Classic S7 communication has no strong protection against someone who can see the network or send messages to port 102.

5.7 Evidence 7: Protocol Evidence Matrix

Question: Complete the matrix for HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7 (Step 10). Fill the "Yours" cells from your own findings.

Protocol	Filter used	Endpoints (source to destination)	Observed operations	Security significance
HTTP (TCP 80)	http.authbasic http.request.method	Yours: _____	Yours: _____	Clear text. Basic authentication is only Base64 encoded, so login material is easy to read. Reveals device web pages.
FTP (TCP 21)	ftp	Yours: _____	Yours: _____	Clear-text commands and login. Banner may reveal product and version.
Modbus (TCP 502)	modbus	Yours: _____	Yours: _____	No authentication or encryption. Write codes can change the process. Registers and values are visible.
DNP3 (port 20000)	dnp3	Yours: _____	Yours: _____	Common in SCADA. Basic DNP3 is unencrypted, so addresses and commands are visible and could be forged.
EtherNet/IP (44818)	enip	Yours: _____	Yours: _____	Device identity (vendor, product, revision) can be read. No default login or encryption.
Siemens S7 (TCP 102)	s7comm	Yours: _____	Yours: _____	Memory reads and writes visible. Device information may be shown. Classic S7 has little protection.

5.8 Evidence 8: Security Assessment and Defensive Monitoring

Question: Write a security assessment with at least five observations, in priority order, and explain what the traffic reveals and how it should be monitored (Step 11).

Check on your evidence: keep an observation only if you really saw it in your capture, and write the figure number in the Evidence column. Delete or change any that you did not see.

No.	Priority	Observation and evidence	Why it is a risk	Recommended control
1	High	Login material sent in clear text or only Base64 encoded (HTTP Basic, FTP). Evidence: Figures 4, 7	Anyone who captures the traffic can read and reuse the credentials to reach devices.	Use HTTPS / SFTP or FTPS, change default passwords, block clear-text services on the ICS network.
2	High	Industrial protocols with no authentication or encryption (Modbus, DNP3, EtherNet/IP, S7). Evidence: Figures 8, 10, 11, 12	Commands and values can be read, and forged commands may change the physical process.	Segment the network, allow only known systems to reach controllers, use secure protocol versions where supported.
3	High	Read or write operations visible (Modbus write codes, S7 Write Var). Evidence: Figures 8, 12	Shows exactly how the process can be controlled and which values can be changed.	Monitor and alert on write and programming operations from unexpected sources.
4	Medium	Device information exposed (FTP banner, EtherNet/IP identity, S7 device data). Evidence: Figures 7, 11, 12	Vendor, model and version let an attacker look for known weaknesses of that device.	Turn off or limit banners, keep an asset inventory, patch, restrict management services.
5	Medium	Many protocols and devices seen from one capture point (excessive network visibility). Evidence: Figure 3	Suggests a flat network with little separation, so an attacker on one device can see all others.	Divide the network into zones (for example by the Purdue model), use firewalls and a DMZ.
6	Medium	Legacy or unneeded services (HTTP and FTP) mixed with control traffic. Evidence: Figures 5, 6, 7	Extra services widen the attack surface and often have weak settings.	Disable unused services and remove clear-text management.

What the traffic reveals

By only listening, I learned which devices exist and how they talk (Figure 3), what they are (banners, identity and device information), which protocols the plant uses, and, where traffic was not protected, the logins and commands. This is very useful to the defender for making an asset inventory, but it is equally useful to an attacker who can see the same traffic.

Comparison of the six protocols

Protocol	Typical port	Typical use	Default login / encryption	Main risk
HTTP	TCP 80	Web pages of devices	Basic login is only encoded; no encryption	Login and pages readable
FTP	TCP 21	File transfer, firmware, configuration	Clear-text login; no encryption	Login and files readable
Modbus TCP	TCP 502	Reading and writing PLC data	None	Anyone can read or write

Protocol	Typical port	Typical use	Default login / encryption	Main risk
DNP3	20000	SCADA to field devices (utilities)	None by default (secure option exists)	Commands can be seen and forged
EtherNet/IP	TCP/UDP 44818, UDP 2222	Rockwell and other CIP devices	None by default (secure option exists)	Device identity exposed; commands forged
Siemens S7	TCP 102	Siemens PLC programming and data	Little or none in classic S7	Memory reads and writes

Defensive monitoring controls

- Use passive monitoring on a network TAP or SPAN port, so the monitoring does not disturb fragile devices.
- Use an ICS-aware intrusion detection system (for example Zeek or Suricata with industrial protocol support) to understand Modbus, DNP3, EtherNet/IP and S7.
- Build a baseline of normal conversations and alert on new devices, new conversations and unexpected sources talking to controllers.
- Alert on write, program-download and stop commands, and on clear-text logins on the ICS network.
- Keep logs and send them to a central place (for example a SIEM) for review.

5.9 Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Base64 is only a way of writing data using text characters. It is not encryption and needs no key, so anyone can reverse it in seconds. The Authorization header carries the user name and password in this encoded form, and it is sent in clear text over HTTP. Anyone who captures the packet can read the credentials.

2. What information can device banners or protocol metadata reveal to an analyst?

They can reveal the vendor, product and model, software or firmware version, device name, serial number, operating system, the services running, and protocol addresses. This helps a defender to build an asset inventory. It also helps an attacker to find known weaknesses of that exact device.

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

Industrial protocols are used between machines to read and control a physical process. They are designed for reliability, speed and simple devices on trusted networks, and often have no login or encryption. Their messages use fixed function codes and register or memory addresses, and a wrong command can affect the physical process. HTTP and FTP are general IT protocols used by people for web pages and file transfer, and they have secure versions such as HTTPS and SFTP.

4. Why is passive packet analysis valuable for understanding an ICS environment?

It sends no traffic, so it cannot disturb fragile control equipment or the process. It shows the real communication: which devices exist, who talks to whom, which protocols and commands are used, and which insecure services are present. This helps to make an inventory and a baseline, and can be done on a live network using a TAP or SPAN port.

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Network segmentation and firewalls that allow only known systems to reach controllers, a DMZ between IT and ICS, ICS-aware monitoring with alerts on write and program commands, encrypted tunnels or VPNs for remote links, strong access control and multi-factor login on remote-access systems, disabling unused services, and good asset and patch management.

7. Conclusion

In this lab I used Wireshark to analyse a supplied ICS packet capture without touching any live system. I identified the communicating systems, examined HTTP and FTP for credentials and device information, and recognised Modbus, DNP3, EtherNet/IP and Siemens S7 traffic and their operations. The traffic showed that many industrial and legacy protocols send information, and sometimes credentials, in clear form with no login or encryption. Passive monitoring is useful to defenders for seeing and protecting the

environment, and the same visibility helps attackers if the network is not segmented and controlled.

8. Recommendations

- Replace clear-text services with secure ones (HTTPS, SFTP or FTPS, SSH) and change all default passwords.
- Segment the ICS network and allow only known systems to talk to controllers.
- Use passive, ICS-aware monitoring with a baseline and alerts for new devices and for write or program commands.
- Limit the information that devices reveal (banners, identity), and keep an up-to-date asset inventory and patching plan.
- Handle captures carefully: work on copies, keep hashes, redact secrets in reports, and never reuse credentials found in traffic.

9. References

International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Lab 07: Exploring ICS Packet Captures (Student Edition). Intro to Critical Infrastructure for IT Learners. <https://www.icdfa.edu.ng>

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